Raffles Institution
Physics Department

Physics Practical
Revision Notes for Year 6 Term 3

This revision notes apply to Prelim and A-Level Practical.

Content **Page**

A GENERAL INFORMATION AND INSTRUCTIONS

H2 Physics 9478/04 (Practical)

1 Format of Paper

Paper number	Total duration	Weighting (%)	Marks
4	2 hr 30 min	20	50

Duration	
First 1 hr 15 min	Either 2 short experiments or 1 long experiment
Second 1 hr 15 min	Change over to the other experiment(s)

Note that the number of experiments (short and long) for 2026 A-Level may or may not be the same as previous years.

2 Marks Allocation (2024 and 2025 A Level Paper 4)

Year	Question	Duration	Topic/Marks
2024	**1**	**1 hr**	**Electrical/14**
	2		**Mechanics/8**
	3	**1 hr**	**Thermal&Mechanics/21**
	4	**30 min**	**Waves/12**
2025	**1**	**1 hr**	**Electrical/12**
	2		**Waves/9**
	3	**1 hr**	**Oscillations/22**
	4	**30 min**	**Waves/12**

Important Note: The format of the Practical Exam you will be sitting for is different from that in 2024 and 2025.

3 What do I have to bring?

3.1 A clear folder that contains your NRIC, student pass or foreign identification document and entry proof (for A-level).

3.2 2-3 blue/black pen, mechanical pencil, lead refills and eraser.

3.3 30 cm long plastic ruler and flexible curve.

3.4 Mathematical set (including protractor, pair of compasses, set-square). Remove the instruction sheet inside the mathematical set.

3.5 Calculators/graphic calculators (extra calculator is recommended) Check SEAB Website for approved calculators list. Learn how to clear memory of graphic calculator.

3.6 Light snacks and drinks (may consume during holding periods for shifts 1 and 3). During your shift, water bottles will be placed along the corridor outside the laboratory.

3.7 This revision notes and other relevant materials (optional, to revise while in holding venue).

3.8 Sweater (optional, to be used in holding venue).

4 What do I have to take note of?

4.1 Check your allocated shift and report to the reporting venue punctually [details to be provided in a separate briefing].

4.2 Handphones may be brought to the reporting venue and surrendered to the Liaison Officers. No other communication or electronic devices should be brought to the reporting venue, e.g. laptops or tablets, smart watches or glasses, thumbdrives, earphones and headphones.

4.3 Pay attention to the instructions from invigilators – **DO NOT START TO WRITE** (e.g. your name) or **TURN OVER THE QUESTION PAPER** or **TOUCH THE APPARATUS** or **LOG IN TO THE LAPTOP** before any instructions are given.

4.4 Once you are in the laboratory, there should be **ABSOLUTELY NO COMMUNICATION** with anyone **BEFORE**, **DURING** and **AFTER THE PAPER**. This includes asking friends to help clear memory of your graphic calculators, borrowing of stationery or hand-shakes, fist-pumps or high-five at the end of the paper.

4.5 **DO NOT CONTINUE TO WRITE** when told by the invigilators during change-over time and at the end of the paper. Turn over your question paper to the cover page. Ask for permission if you want to write your names at the end of the paper.

4.6 **DO NOT TOUCH THE APPARATUS** during the the change-over time.

4.7 Irregularity report will be filed if candidate fails to abide by any instructions from the invigilators.

4.8 Plan and use your time wisely. Make sure you read and understand the entire experiment before setting up your experiment. As a guide, you should spend a maximum of 15 minutes for data collection for each short experiment and a maximum of 40 minutes for long experiment (this include data in different parts of the question). In the worst-case scenario, you must collect all the necessary data before change-over.

4.9 If you complete the report for first experiment(s) (before change-over), you are free to use the remaining time to read and prepare for second experiment(s).

4.10 Check that you have completed every part for all experiments. Any omission will result in a loss of mark(s).

4.11 At the change-over and the end of the paper, follow the instructions from the invigilators/Subject Supervisor on restoring the apparatus back to its original state. Do not damage any apparatus. Check that you do not bring any of the apparatus out of the lab as this will be considered as cheating.

4.12 Queue up quietly outside the laboratory. Make your way quickly to the holding area when instructed.

4.13 Refrain from discussing the practical. Just be glad that you have done your best :)

5 What if I am unable to proceed with the experiment?

5.1 Raise your hand to request for help if you are unable to set up your apparatus **15 MIN AFTER START OF PAPER**. You may only ask for help **RELATED** to the **EXPERIMENTAL SET-UP**. No assistance may be given for performing the experiment, interpreting the question, usage of any functions in Excel or writing the report. Any assistance rendered will be reported and marks may, or may not, be deducted.

5.2 Be specific in your request for help so that only the necessary assistance is rendered. Vague request will result in more time lost.

For example, instead of saying “I don’t know what’s wrong with my set-up.”, you should say “Please help me check my circuit as I am not able to get any readings on my DMM.” or “I need help to stop the plasticine from slipping off the wire.”.

Neither should you ask questions like “How do I oscillate this plasticine?” or “How do I set up this circuit?”.

5.3 You may request the Subject Supervisor **AT ANY TIME** to check your instrument if you suspect it to be faulty. Raise your hand to alert the invigilator. No marks will be deducted.

5.4 You may also request for trouble-shooting if you encounter any operational difficulties with your laptop.

B PRACTICAL SKILLS

1 Recording and Processing

1.1 Do not use correction tape or fluid. Just cancel neatly and write again.

1.2 Use blue/black ink for all readings and workings.

1.3 Use pencil for graph work, diagrams and drawing of table outline.

1.4 Always linearise the equation first.

Some equations are already linearised. You just need to recognise the quantities to plot. **DO NOT WASTE TIME** rearranging the terms.

Write a statement to show the quantities plotted, the gradient and vertical-intercept.

1.5 Uncertainty in a measurement

$\Delta x = \frac{1}{2}(x_{max} - x_{min})$ or 3 × least count of instrument, whichever is greater.

For timing experiment, Δt typically ranges from 0.1 s to 0.5 s. There may also be exceptions due to the difficulty of taking a time measurement. **[e.g. Mock 1 2019/Q3]**

For some measurements that are inherently more difficult to take, the uncertainty in different measurements with the same instrument may differ. **[e.g. WS20 2023/Q3]**

If X can be measured to greater precision than Y, then $\Delta X < \Delta Y$.

Always be mindful of the possible sources of uncertainties when doing the experiment.

1.6 Absolute uncertainty: 1 s.f.; Percentage uncertainty: 2 s.f.

1.7 Record all readings in a table.

1.8 Include units for all column headings in the table.

1.9 Always include the first set of reading (usually taken prior to tabulation) in your table.

1.10 Straight line: at least 6 readings; at least 8 readings for non-linear graph.

Take more readings around turning points, if time permits.

1.11 Present raw data to the same number of d.p. as instrument resolution.

1.12 Processed data has the same (least s.f. or d.p. if raw data has differing s.f. or d.p.) number of s.f. (product/quotient) or d.p. (sum/difference) as the raw data. If processed data is 1 or 2 s.f., you may add 1 more s.f. to every value in the column.

1.13 Take repeated readings and calculate average for measurements that may not be uniform or cannot be measured precisely due to random errors (e.g. loading/unloading experiment, rebound height of a ball, diameter of a wire, timing of oscillations e.t.c.).

If unsure, repeat if time permits.

1.14 For oscillatory experiments, time for N oscillations should be at least 20 s.

For severely damped oscillations, obtain 3 sets of timing if time permits.

2 Excel & Graphing Skills

Below are the functions in Excel that you are required to know:

2.1 Perform the following mathematical operations:

- calculating **sines, cosines, tangent** (and their inverse functions) on **angles in radians**(**converting from degrees** as necessary)
- calculating **exponentials** and **logarithms**
- using **scientific notation** (e.g. 3.00E+14 for 3.00×10^{14})

2.2 Use features for numerical data manipulation

- use and input formulae and duplicate formulae across cells
- adjust formatting (e.g. data type) and duplicate formatting across cells

2.3 Plot a labelled 2D scatter plot using selected data from a table:

- adjust the scale of both axes
- use selected data points to give a **line of best fit** (e.g. add linear trendline function)
- display the **equation of a trendline**
- determine **area under a curve (e.g.** using trapezium rule)
- determine the **gradient at a point** on a curve (e.g. calculating in a small interval near a point using SLOPE function)

General guide to steps needed to process your data in Excel

2.4 Construct the data table in Excel with column headings.

2.5 Plot appropriate graph using Insert -> Chart -> Scatter.

2.6 Insert trendline by right clicking on the plot. Format trendline to insert equation of line.

2.7 Sketch graph on question paper with appropriately labelled axes, origin and vertical intercept of graph shown. Write equation of line on the sketch.

Data-Based Question

2.8 For processing of large data sets in data-based question, do an initial plot to identify and remove anomalous data points.

2.9 Use SLOPE function to calculate gradient at a point for the data set if required.

2.10 Use trapezium rule to calculate the area under the graph if required.

2.11 Linearisation of equation, plotting additional graphs and removal of more anomalous data depending on the requirements of the question.

See Practical Revision Videos for a walkthrough of common Excel functions required. Screenshots of the steps can be found in the Appendix from page 13 onwards.

3 Gradient, Vertical-Intercept & Unknown Constants

3.1 Obtain the gradient and vertical intercept (to 3 s.f.) from the equation of the line in Excel.

3.2 Calculate the unknown <u>constants</u> separately and include any <u>units</u>.

4 Anomaly, Significant Errors and Improvements

4.1 Identify and remove anomalous data (if any) based on your plot and indicate in your table in the question paper the anomalous point that was removed.

4.2 For identification of anomalous point(s) in the given big data set (downloaded file during exam) you can do an initial plot of the given data set. Remove these anomalous data points from the data set.

4.3 Sources of significant errors can arise from measuring instruments or experimental techniques. Errors should have significant impact on the readings and are inherent (cannot be avoided given the apparatus and technique) and specific to the experiment. General errors such as parallax errors are not accepted.

For errors arising from measuring instruments, propose more precise instruments.

For errors arising from experimental techniques, propose better approach.

State the error and the quantity that is affected.

> "There is uncertainty in determining the centre of the bob. This results in an uncertainty in the measurement of the pendulum length *l*."

Improvements may involve use of other commonly found laboratory apparatus.

> "Set up a pin below the bob as a fiducial marker to aid in starting and stopping the stopwatch. This would improve the accuracy of the timing."

Improvements should not be procedures or good practices already given in the question or expected in the experiment.

5 Other Discussions

5.1 Justify the number of significant figures given in your values. **[Mock 1 2019/Q3]**

> Given equation: $t = \frac{k}{y}$
>
> Since values of *y* are 2 s.f., and values of *t* are 4 s.f., *k* follows the least s.f. which is 2 s.f.

Note that you should mention the number of significant figures of all quantities in the equation before coming to a conclusion on the number of significant figures for the required quantity based on the principle of following the least number of significant figures.

5.2 State whether the results of your experiment support the suggested relationship.

Justify your conclusion by referring to your answer in **(a)(ii)**. **[Mock 1 2019/Q3]**

> $k_1 = 47$ cm s, $k_2 = 55$ cm s
>
> Percentage uncertainty in $k = \frac{\frac{1}{2}(k_2 - k_1)}{\frac{1}{2}\langle k_2 + k_1 \rangle} \times 100\% = \frac{\frac{1}{2}(55-47)}{\frac{1}{2}(55+47)} \times 100\% = 7.8\%$.
>
> Since this is larger than the percentage uncertainty of 2.3% in **(a)(ii)**, the results do not support the relationship.

OR

(If percentage uncertainty in k is calculated to be smaller, e.g. 1.8%)

Since this is smaller than the percentage uncertainty of 2.3 % in **(a)(ii)**, the results supports the relationship.

5.3 Show whether your value in **(a)(iii)** explains the difference between your two values of *X*. **[WS20 2023/Q3(d)]**

The use of an electronic balance to determine the mass of the metre rule gives a more accurate result. Thus we calculate the percentage difference of the experimental result in **(c)** with this direct measurement in **(d)** as the base value.

$X_1 = 57.5\text{ g}$ (calculated), $X_2 = 68.03\text{ g}$ (measured with balance)

Percentage difference in $X = \dfrac{68.03 - 57.5}{68.03} \times 100\% = \dfrac{10.53}{68.03} \times 100\% = 15\%$.

Since the percentage difference in X (15%) is greater than the percentage uncertainty in h/L (3.9%), **(a)(iii)** does not explain the difference between the two values of *X*.

5.4 Use your results to estimate a value of *d* where the value of *T* is the same with and without the mass. **[WS10 2021/Q3(e)]**

Always <u>present your results</u> in a <u>table</u>.

For each set of data, determine the value of *T* with and without the mass.

Plot a graph in Excel to plot *d* vs ΔT and obtain the equation of the line. The vertical intercept gives the value of *d* where ΔT = 0.

6 PLANNING QUESTION(S)

The planning question will have a weighting of around 10 marks in Paper 4.
You can either encounter a couple of short planning questions or a long planning question.

Report Writing For Planning Question

The information given at the start of the question provides context to the question. Such information may or may not be useful in answering the question. There is usually more than 1 way of answering a planning question and the marking scheme varies from year to year. In general, your answer to a **long planning question** (~10 marks) should include the following 4 sections:

1 Diagram

- Draw clearly labelled 2D diagram(s) – top, front and/or side view that show the relative positions of the apparatus in workable arrangements.
- Apparatus should be supported by clamps and stands or placed on the bench or floor (represented by a line). Apparatus should not be hanging in the air.

- Unstable apparatus (such as motor, pulley, retort stands) should be firmly clamped to the bench.
- Conventional circuit symbols must be used for circuit diagrams.
- You may use some or all the apparatus suggested in the question, as well as any other equipment usually found in a Physics laboratory.

2 Procedure

The procedure must be logically sequenced and clearly numbered. Leave a line between each step for amendments. Students may use any typical apparatus found in the physics laboratory. The procedure should include:

- additional descriptions of the setup, if the diagrams cannot fully explain the intentions.
- definitions of all symbols used.
- how the independent and dependent variables are measured using the correct instruments and methods.
- equations used to determine the independent and dependent variables, if they cannot be measured directly.
- methods to vary the independent variable.
- values that are not given must be measured and not assumed (e.g., cannot assume that diameter of wire is uniform, temperature of ice is $0°C$, etc).
- steps to be repeated to obtain 8 sets of data.
- methods to measure and keep constant the other variables (preferably measurable) for a valid experiment.

Reliability

Include precautions to improve the reliability (i.e., accuracy and precision) of the experiment. These may include:

- crucial arrangement of apparatus.
- calibration of instrument.
- steps to reduce / eliminate sources of error.
- taking preliminary readings to determine a suitable range of the independent variable which will produce a valid range of dependent variable (specific values or trends are not required).

3 Analysis

Describe how the data obtained should be processed to achieve the objective(s) of the experiment.

Case 1: If relationship is given or known (from the syllabus)

- Linearise the equation, if necessary.
- State the variables to be plotted.
- Explain how a straight line graph verifies the relationship and how the gradient and vertical intercept can be used to determine unknown constants.

Given the equation: $I = I_0 e^{-\mu t}$ where μ and I_0 are constants.

Linearise equation: $\ln(I) = \ln(I_0) - \mu t$

Plot a graph of $\ln(I)$ against *t*.

If the relationship is true, a straight line graph will be obtained, where the gradient = $-\mu$ and the vertical intercept = $\ln(I_0)$.

$I_0 = e^c$ where *c* is the vertical intercept.

Case 2: If relationship is NOT given and NOT known

Assumed equation: $y = kx^n$ where *k* and *n* are constants.

Linearise equation: $\lg(y) = n\lg(x) + \lg(k)$

Plot a graph of $\lg(y)$ against $\lg(x)$.

If the relationship is true, a straight line graph will be obtained where the gradient = *n* and the vertical intercept = $\lg(k)$.

$k = 10^c$ where *c* is the vertical intercept

4 **Safety Precautions**

- State reasonable hazards that are specific to the experiment AND how to avoid them.
- State the actions to be taken, NOT the actions to avoid. For example, to prevent scalding, “wear glove when handling the hot beaker” rather than “do not handle the hot beaker with bare hands”.
- Rationale must be provided for the precaution taken.
- Hazards can be related to the apparatus (e.g., high-voltage supply), the materials (e.g., hot oil) or the procedure (e.g., lifting a heavy load).

Short Planning Question

The focus for the short planning question is to answer the points stated in the question. If you are asked to verify a relationship **[e.g. WS12 2025/Q3(h)]**, ensure you answer the question in your analysis. Typically you have around 15 min for a question of 5 marks so you cannot afford to write to the level of detail as that of the long planning question.

C APPARATUS

1 Resolution of Common Apparatus (d.p. of raw data)

Meter Ruler	1 mm, **0.1 cm**, **0.001 m** (preference in **bold**)
Digital Vernier Calipers	**0.01 mm**, **0.001 cm**, 0.00001 m
Digital Micrometer Screw-Gauge	**0.001 mm**, 0.0001 cm, 0.000001 m
Protractor	1 degree
Stopwatch	0.01 s
Thermometer	½ smallest division
Measuring Cylinder	½ smallest division
Digital Multi-Meter (DMM)	Record to number of d.p. shown on screen
Slotted Mass	Record value as stated
Resistor	Record value as stated

2 Tying of Knots

2.1 Purchase some thick cotton string to practise tying of slip (adjustable) and dead knots.

Slip Knot

Dead Knot

3 Digital Vernier Calipers & Micrometer Screw Gauge

3.1 The digital vernier calipers is a versatile length-measuring instrument. It can be used to measure the dimensions of an object, the diameter or the depth of a hole.

The range is typically 150 mm. Always close jaws and zero before using.

Record what is shown on the display.

3.2 The micrometer screw gauge is a high precision instrument used for measuring the dimension of a small object such as the diameter of a ball bearing or a wire.

The range of measurement is typically 25 mm. Always close jaws and zero before using.

Record what is shown on the display.

4 Digital Multimeter (DMM)

4.1 A multimeter is used to measure current, potential difference (voltage) and resistance in an electrical circuit.

4.2 Turn the dial to select the appropriate range and type (d.c. or a.c.) of measurement.

4.3 For current measurement, press SEL/Δ (yellow button) to toggle between AC or DC.

4.4 Connect 4 mm plug wires into the ground terminal and another appropriate terminal (see labels above terminals).

4.5 Use the same setting throughout the experiment. For example, do not change the setting from mA to A midway through the experiment.

4.6 Readings may fluctuate due to high sensitivity or poorly connected circuit. In such cases, wait a few seconds and just take the reading.

4.7 Check that the letter **H** (hold) does not appear on screen in which case, the reading will not change. Press HOLD (blue button) to turn off the function.

4.8 If DMM turns off during the experiment, just press SEL/Δ (yellow button) to turn it on.

4.9 Ignore the negative sign when measuring DC current or p.d. or simply swap the connectors to the terminals.

4.10 If DMM does not register a reading, check that

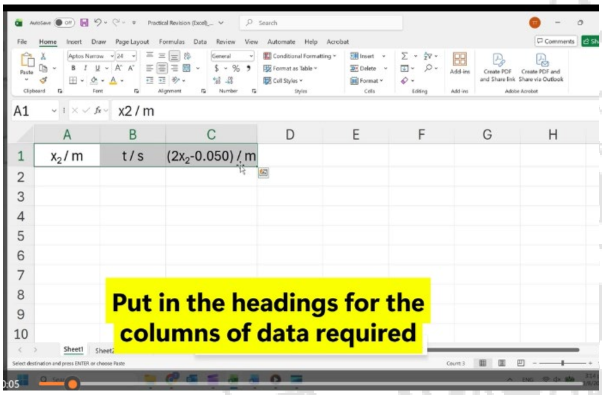
- the circuit is connected correctly,
- all connecting wires are connected securely to components,
- the correct setting and terminal connections are used

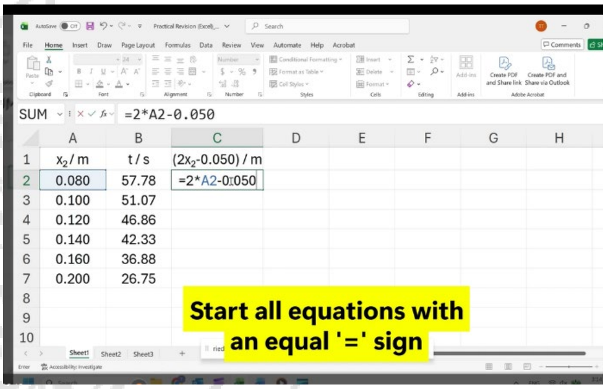
Otherwise, assume DMM is faulty. Alert invigilator immediately.

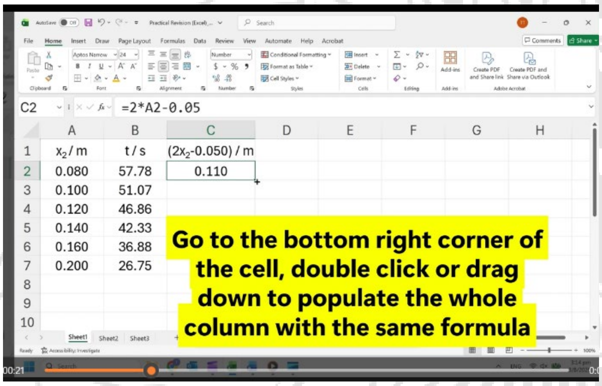
4.11 If the battery is weak (a battery icon appears on screen), alert invigilator immediately.

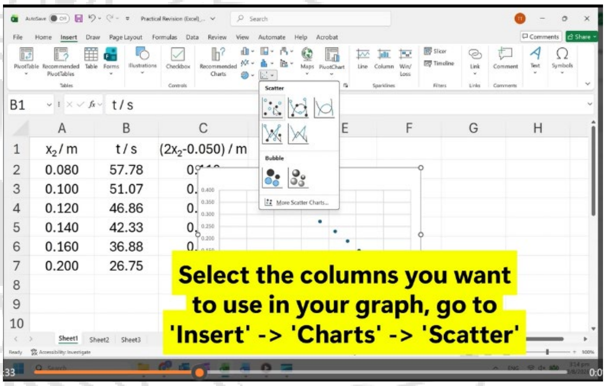
D APPENDIX

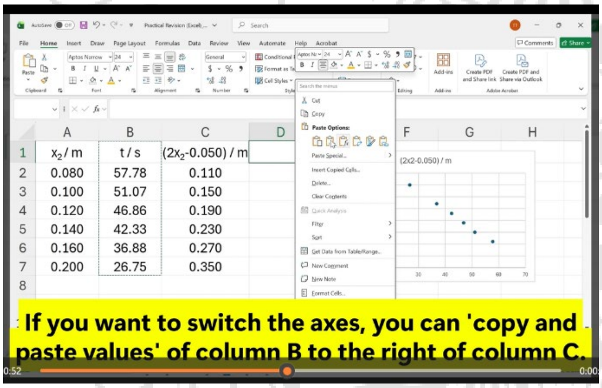
Setting up of table and graph plotting

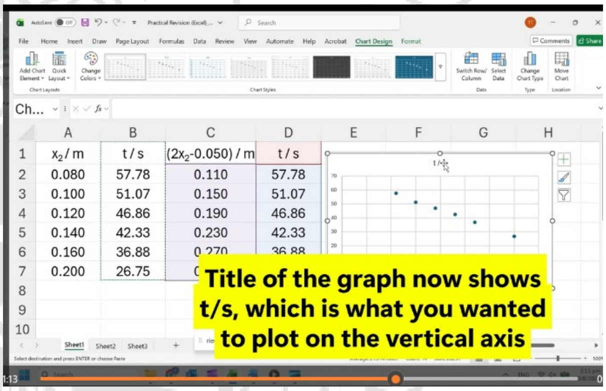


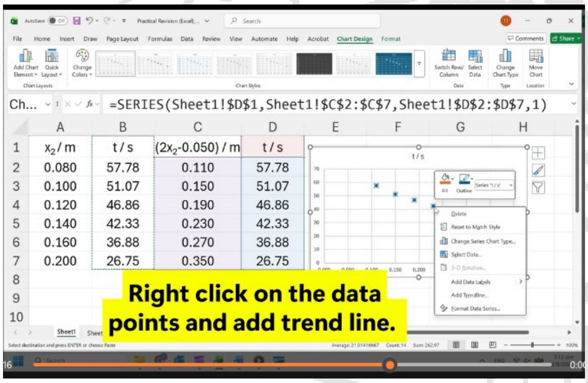


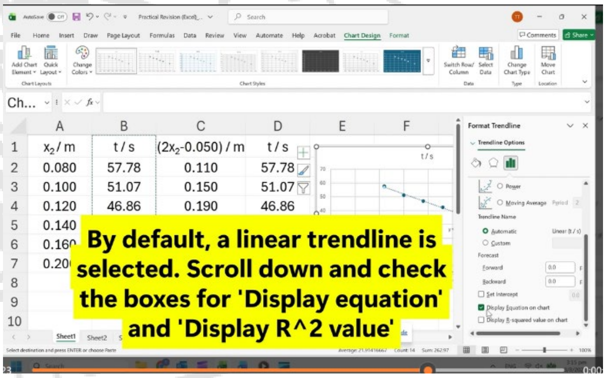


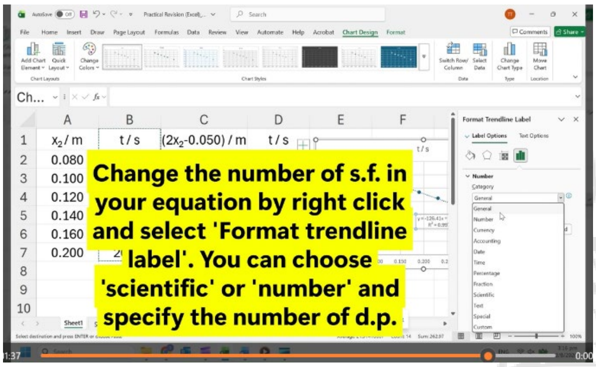




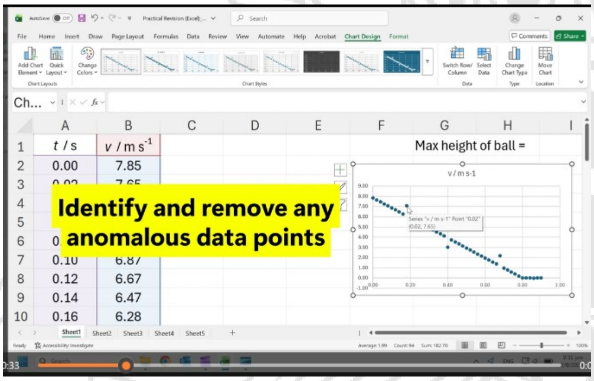


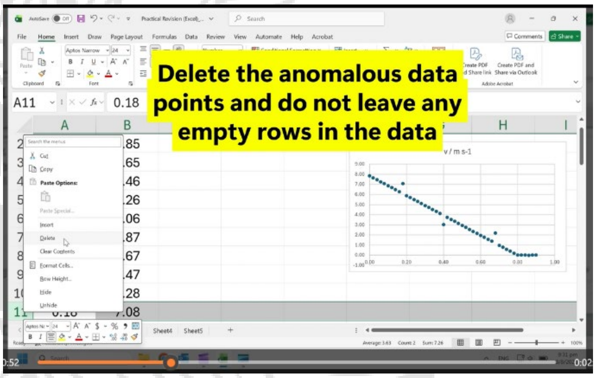


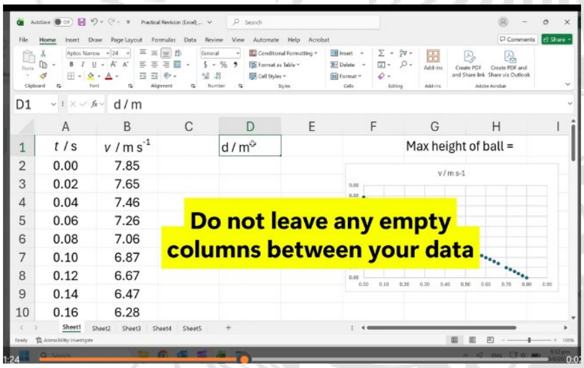




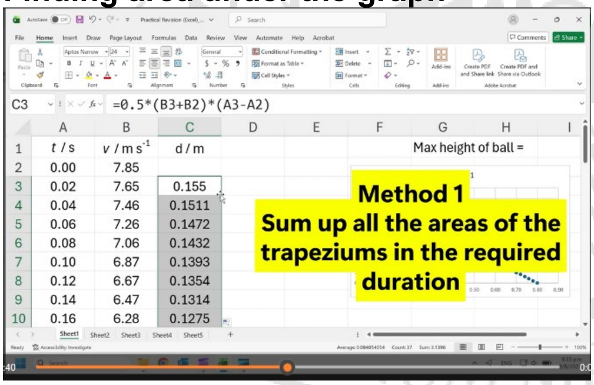
Tackling data-based question

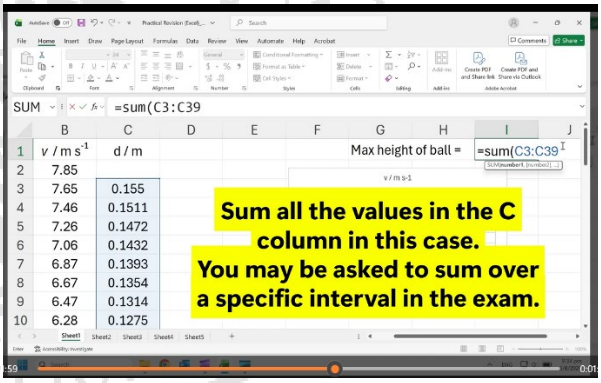


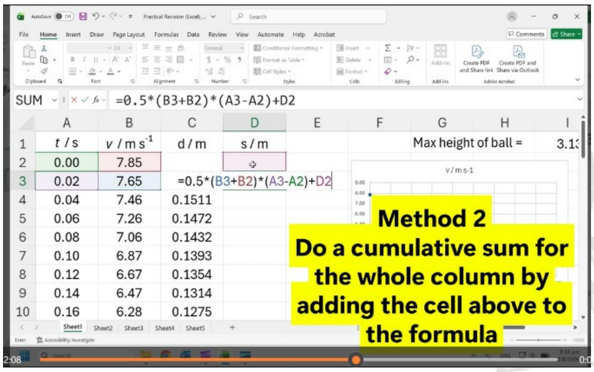


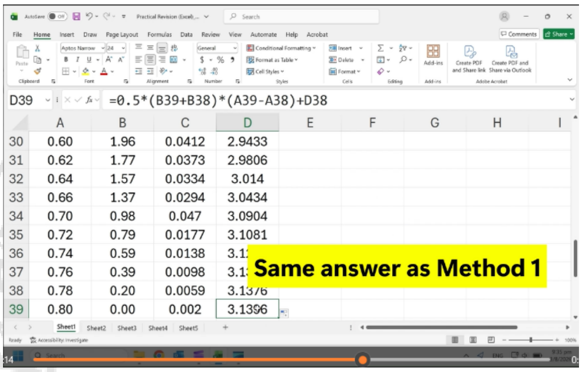


Finding area under the graph

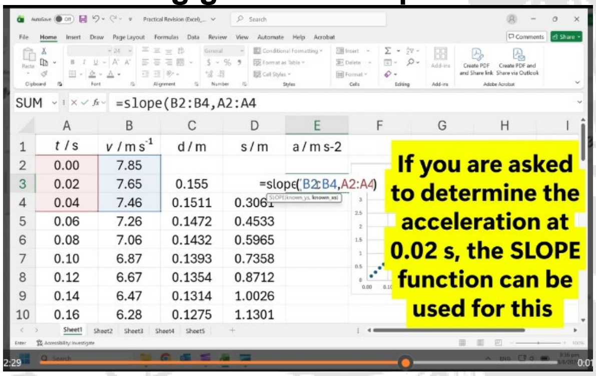








Determining gradient at a point



Changing the scale of an axis